

KYAW SWAR HEIN

DATA ENGINEER



CONTACT

📞 09 457 151 081

✉️ heinkyawswar804@gmail.com

🌐 <https://ksh-space-portfolio-next.vercel.app/>

🐙 <https://github.com/Kyawswar-hein>

📍 Hlaing, Yangon

SKILLS AND EXPERTISE

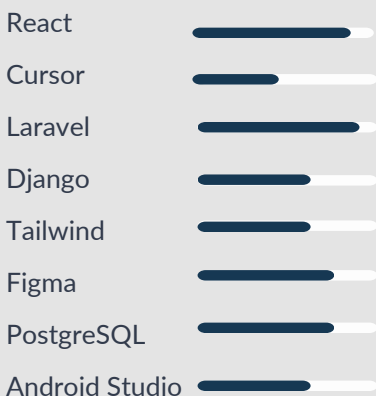
Operating Systems



PROGRAMMING LANGUAGES



FRAMEWORKS AND TOOLS



PROFILE

Engineering Graduate specializing in AI and Data Analytics with a proven track record in full-stack software development. From architecting Spring Boot and MERN-based systems to deploying image classification models, I bridge the gap between robust backend engineering and intelligent automation. I am passionate about building scalable AI solutions and workflow automations that solve real-world operational challenges.



EDUCATION

Myanmar Institute of Information Technology 2017 - 2026

- B.E. (Hons) | Electrical Communication Engineering



WORK EXPERIENCE

DIR Ace Technology Company Ltd

2025 DEC - 2026 MAR

Fullstack Developer Intern

- Developed a Full-Stack Review Count Management System using Java Spring Boot and PostgreSQL to automate the tracking of project quality metrics (DQI, CQI, TQI).
- Architected a Role-Based Access Control (RBAC) system using Spring Security, implementing custom filters and success handlers for Admin, Editor, and Viewer roles.
- Data Import Module using Apache POI, enabling bulk processing of Excel datasets with transactional integrity and validation logic.
- Designed an Interactive Analytics Dashboard using Chart.js and Thymeleaf, providing real-time visual comparisons of "Base Week" vs. "Compare Week" performance.

Star88 Trading Company Ltd

2022 MAY - 2022 NOV

Bridge It Support (Project Based)

- Set up different devices, Barcode Scanners, printers for POS (point of sales) and ERP (enterprise resource planning features) of Retail and Wholesale Mart.
- Acted as bridge between the client company and the two software companies.
- Data cleaning, processing for over twenty thousand different items with price setting and barcode management then integrated into ERP software.

CLOUD SERVICES

AWS (EC2, Lightsail)
Google Cloud
Azure

LANGUAGES

Burmese (Native)
English (Business level)

HOBBIES & INTERESTS

Solving puzzles
Watching Sci-Fi movies
Listening to music
Graphic design

FAVORITE QUOTATION

"Learning gives you the tools, but creation is what sharpens them."



PROJECT EXPERIENCES

3D Space-Themed Portfolio Website

(Next.js, Three.js, Framer Motion, TypeScript)

- Developed a visually immersive portfolio by integrating Three.js to render interactive 3D space-themed models and backgrounds.
- Engineered a dynamic and fluid user experience using Framer Motion for complex animations and smooth page transitions across all components.
- Built a fully responsive, multi-section website with Next.js 13 and TypeScript, and deployed it on Vercel for optimized performance and availability.

E-Grocer - Android Application

(Kotlin, Glide, Firebase)

- Developed a native Android e-commerce app in Kotlin, featuring a complete user flow from product browsing to checkout.
- Implemented the MVVM architecture, using Retrofit for API communication and the Room database for offline data caching.
- Integrated Firebase for secure user authentication, cloud data storage (Firestore), and media management (Cloud Storage).

Student Management System -SMS

(Django, Tailwind, MySQL, SQLite)

- Designed and implemented a relational database schema in MySQL to efficiently manage student, course, and academic records.
- Developed a role-based web application with distinct user interfaces and permissions for Admins, Faculty, and Students.
- Automated key academic processes, including user management, course registration, and grade reporting, with custom logic for complex student requirements.

Music Genre Classification - Machine Learning Webapp

(Django, Scikit-learn, Librosa)

- Engineered a machine learning pipeline to classify music genres by extracting audio features from .wav files using the Librosa library.
- Trained a Random Forest Classifier model on the extracted features and serialized it for production use.
- Built and deployed a user-friendly web interface with Django that allows users to upload audio files and receive instant genre predictions.



Weapon Detection & Classification System

(TensorFlow, Keras, Scikit-Learn, NumPy, Matplotlib & Seaborn)

- Developed an end-to-end computer vision pipeline to identify and classify security threats (Knife, Longgun, Shotgun).
- Performed a comparative analysis between VGG16 and InceptionResNetV2 architectures using Transfer Learning to optimize for classification accuracy and model depth.
- Implemented InceptionResNetV2 with custom global average pooling and dropout layers to handle complex feature extraction from 299x299 image inputs.
- Utilized ImageDataGenerator for real-time data augmentation (rotation, zooming, and horizontal flipping) to improve model generalization and robustness against varied camera angles.
- Evaluated performance using Confusion Matrices and Seaborn visualization to identify and reduce false-positive rates between similar weapon categories.

ABOUT ME

Passionate about building with technology, I see challenges as opportunities for growth. I am a dedicated learner, ready to apply my problem-solving skills and contribute to innovative solutions in a professional environment.